SC Algebra 1

Reteach: Intercepts and Writing Intervals

Name: _____

Part A — Guided: Finding x- and y-intercepts

Key Idea

x-intercept: where the graph crosses the _____. On the x-axis, y is ALWAYS ___. Write it as (x , 0).

y-intercept: where the graph crosses the _____. On the y-axis, x is ALWAYS ___. Write it as (0 , y).

To find the x-intercept, make y = 0. To find the y-intercept, make x = 0.

Example 1: 2x + 3y = 12

"Cover-up" trick for Ax + By = C

x-intercept: make y = 0 → 2x + 3(0) = 12

2x = ____ → x = ____ → (___ , 0)

y-intercept: make x = 0 → 2(0) + 3y = 12

3y = ____ → y = ____ → (0 , ___)

Example 2: y = 2x − 6

Slope-intercept form: the y-intercept is b

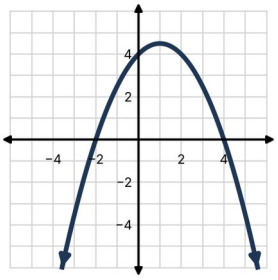
x-intercept: make y = 0 → 0 = 2x − 6

6 = 2x → x = ____ → (___ , 0)

y-intercept: make x = 0 → y = 2(0) − 6

y = ____ → (0 , ___)

Example 3: From a graph



Where does it cross the x-axis? _____

Where does it cross the y-axis? _____

A graph can have more than one x-intercept.

Example 4: From a table

x	−4	−2	0	2	4
y	−3	0	3	6	9

x-intercept: find the row where **y = 0**: _____

y-intercept: find the row where **x = 0**: _____

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Part A — Practice: Finding x- and y-intercepts

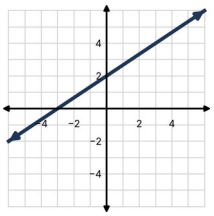
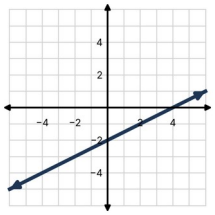
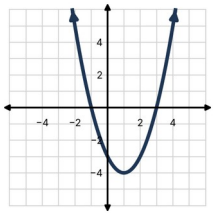
A1. Standard form (use the cover-up trick)

1. $x + 2y = 8$ x-int: _____ y-int: _____	**2.** $3x + y = 9$ x-int: _____ y-int: _____	**3.** $4x - 2y = 16$ x-int: _____ y-int: _____
4. $5x + 2y = -10$ x-int: _____ y-int: _____	**5.** $-3x + 4y = 12$ x-int: _____ y-int: _____	**6.** $6x - 5y = 30$ x-int: _____ y-int: _____
7. $2x - 7y = 14$ x-int: _____ y-int: _____	**8.** $-x - 3y = 6$ x-int: _____ y-int: _____	**9.** $3x + 4y = 6$ ★ x-int: _____ y-int: _____

A2. Slope-intercept form

10. $y = 3x + 12$ x-int: _____ y-int: _____	**11.** $y = -2x + 8$ x-int: _____ y-int: _____	**12.** $y = \frac{1}{2}x - 3$ x-int: _____ y-int: _____	**13.** $y = -4x - 10$ x-int: _____ y-int: _____

A3. From a graph

14.  x-int: _____ y-int: _____	**15.**  x-int: _____ y-int: _____	**16.**  x-int: _____ y-int: _____

A4. From a table A5. In context

17.

x	−2	−1	0	1
y	9	6	3	0

x-int: _____
y-int: _____

18.

x	−6	−3	0	3
y	−2	0	2	4

x-int: _____
y-int: _____

19. A gift card has $B = 60 - 4s$ dollars left after s smoothies. Find both intercepts. What does each mean?

20. A diver's depth is $d = 5t - 40$ feet after t seconds. Find both intercepts. What does each mean?

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Part B — Guided: What is an interval?

Speak Like A Mathematician

interval: a _____ of the number line: all the numbers between two boundary points, or all the numbers going on _____ in one direction.

boundary point: the number where an interval _____ or _____.

open circle: the boundary point itself is _____ included. We use < and >.

Reading the symbols

Symbol	Say it	On the number line
x < 3	x is less than 3	every number to the LEFT of 3
x > 3	x is greater than 3	every number to the RIGHT of 3

Trick: the pointy end of < always points at the SMALLER number. x < 3 means x is smaller than 3.

Words → inequality

In words	As an inequality
all numbers to the LEFT of −2	_____
all numbers to the RIGHT of 1	_____
all numbers BETWEEN −1 and 4	_____
all numbers LEFT of −3, and all numbers RIGHT of 2	_____

Is it in the interval?

The interval −2 < x < 5



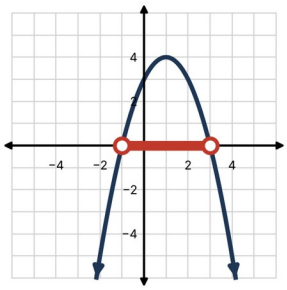
x = 0 IN / OUT

x = 5 IN / OUT

x = −3 IN / OUT

x = 4.5 IN / OUT

On a graph, an interval is a stretch of the x-axis.



The graph is ABOVE the x-axis over the red stretch.
That stretch is the interval _____

Write in words: x > −4 means _____

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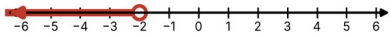
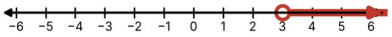
Part B — Guided: Writing intervals with inequalities

Key Idea

"Where" questions (positive, negative, increasing, decreasing) are answered with ___-values only.

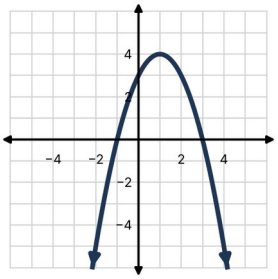
Read the graph from _____ to _____, like a sentence. Use the x-values where the graph turns or crosses the x-axis.

When x is BETWEEN two numbers, the _____ number goes on the left: $-2 < x < 5$

Everything to the LEFT	Everything to the RIGHT	BETWEEN two numbers
		
$x < -2$	**$x > 3$**	**$-1 < x < 4$**

Two separate pieces? Write both and join them with "and": $x < -1$ and $x > 4$

Example 1: Positive and negative



Step 1: Mark the x-intercepts: x = ____ and x = ____

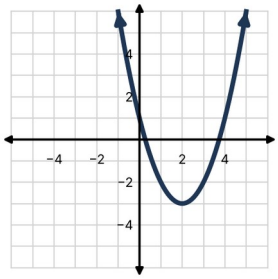
Step 2: Where is the graph ABOVE the x-axis?

Positive: _____

Step 3: Where is it BELOW?

Negative: _____

Example 2: Increasing and decreasing



Step 1: Where does the graph turn? x = ____

Step 2: Trace left to right. Going DOWN first, then UP.

Decreasing: _____

Increasing: _____

Use the x-value of the turning point, not the y-value.

Watch out

× Using y-values. The answer is about WHERE on the x-axis.

× Writing $5 < x < -2$. The smaller number always goes first: $-2 < x < 5$.

× Writing $-1 > x > 3$ for two pieces. Two pieces need two inequalities: $x < -1$ and $x > 3$.

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Part B — Practice: Writing intervals with inequalities

B1. Write the inequality for each shaded number line

1.



_____

2.



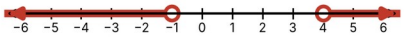
_____

3.



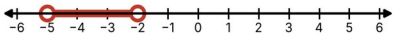
_____

4.



_____

5.



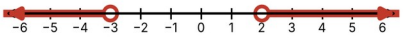
_____

6.



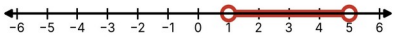
_____

7.



_____

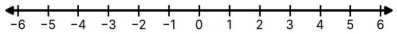
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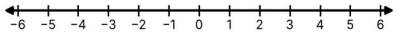
_____

B2. Shade the number line

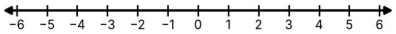
9. x < 4



10. −2 < x < 3

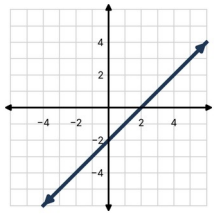


11. x < −4 and x > 1



B3. Where is each function positive? Negative?

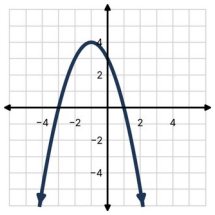
12.



pos: _____

neg: _____

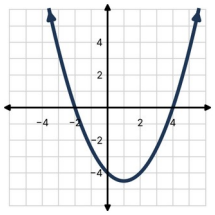
13.



pos: _____

neg: _____

14.

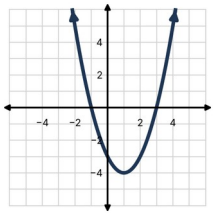


pos: _____

neg: _____

B4. Where is each function increasing? Decreasing? (#18: positive and negative)

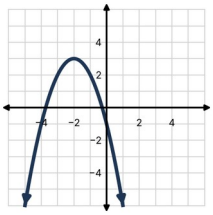
15.



inc: _____

dec: _____

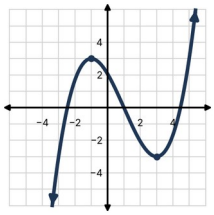
16.



inc: _____

dec: _____

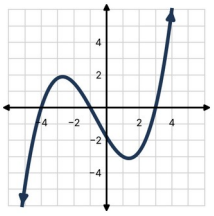
17.



inc: _____

dec: _____

18. ★ Positive? Negative?



pos: _____

neg: _____